

3070. Count Submatrices with Top-Left Element and Sum Less Than k

Difficulty: Medium

Link: <https://leetcode.com/problems/count-submatrices-with-top-left-element-and-sum-less-than-k/description/?envType=daily-question&envId=2026-03-18>

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Problem:

You are given a **0-indexed** integer matrix `grid` and an integer `k`.

Return the **number** of submatrices that contain the top-left element of the `grid`, and have a sum less than or equal to `k`.

Example 1:

7	6	3
6	6	1

7	6	3
6	6	1

7	6	3
6	6	1

7	6	3
6	6	1

Input: grid = [[7,6,3],[6,6,1]], k = 18

Output: 4

Explanation: There are only 4 submatrices, shown in the image above, that contain the top-left element of grid, and have a sum less than or equal to 18.

Example 2:

7	2	9
1	5	0
2	6	6

7	2	9
1	5	0
2	6	6

7	2	9
1	5	0
2	6	6

7	2	9
1	5	0
2	6	6

7	2	9
1	5	0
2	6	6

7	2	9
1	5	0
2	6	6

Input: grid = [[7,2,9],[1,5,0],[2,6,6]], k = 20

Output: 6

Explanation: There are only 6 submatrices, shown in the image above, that contain the top-left element of grid, and have a sum less than or equal to 20.

Constraints:

- m == grid.length
- n == grid[i].length
- 1 <= n, m <= 1000

- $0 \leq \text{grid}[i][j] \leq 1000$
- $1 \leq k \leq 10^9$